Multimodal Fusion of Histopathology and Genomics for Breast Cancer Molecular Subtyping

Individual Project Pitch

Base Project: Kamineni - Multimodal Breast Cancer Diagnostic & Outcome Assistant

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Problem Statement

Clinical Context. Breast cancer information is scattered across pathology, radiology, clinical history, biopsy, treatment and follow-up records (Kamineni brief).

The Gap. Molecular subtype (PAM50) and receptor status (ER, PR, HER2) drive treatment selection but establishing them requires IHC/molecular assays that are expensive, slow, and unevenly available.

A routine **H&E slide already exists for every patient.**

Clinical Question. How much molecular information is recoverable from routine histopathology alone and how much does adding transcriptomics improve it?

Scope.

In scope: predicting ER, PR, HER2 status and PAM50 subtype from whole-slide images (WSI) + RNA-seq.

Not in scope: replacing IHC; autonomous diagnosis.

Workflow Fit (Kamineni).

Pathology/Radiology $\rightarrow$ Clinical review / MDT $\rightarrow$ Treatment planning $\rightarrow$ Follow-up

Data Availability

Hospital Data (Kamineni). Status: **Awaiting.**

Public Fallback **Primary: TCGA-BRCA** (∼1,098 cases) via the GDC Data Portal.

- Modalities: H&E WSI, RNA-seq, CNV, methylation, miRNA, clinical
- Labels *already present*: PAM50 subtype, ER/PR/HER2 status derived from TCGA consortium assays, **no manual annotation required**
- **Open-access tier** covers everything needed (WSI, expression, clinical); only low-level germline/BAM data requires dbGaP authorization

External cohort: CPTAC-BRCA (∼134 patients) - adds proteomics, freely public via the Proteomic Data Commons; enables genuine cross-cohort testing.

Preprocessing Tooling.
Macenko stain normalization · CLAM/QuPath tiling · ComBat batch correction · RSEM quantification

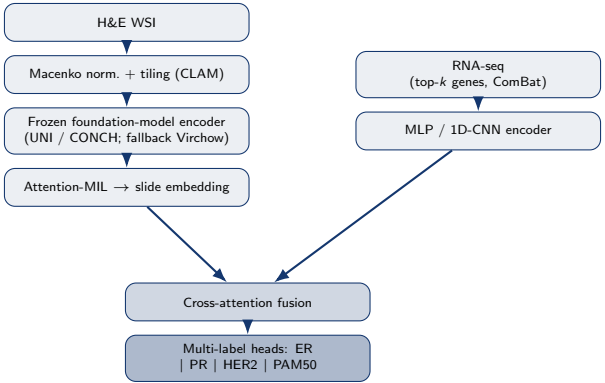
Secondary Fallback (no omics).
CBIS-DDSM (1,566 pts., mammography) · BreaKHis (7,909 images, 82 pts.) - fully open, no credentialing

State of the Art/Relevant Work

Work	Data	Result
M3FusionNet (Shah et al., 2026)	TCGA-BRCA $n\approx1036$ + CPTAC $n\approx134$; ResNet50 feature-level fusion	In-dist. AUC 0.99 / 0.96 / 0.98 / 0.96 (ER/PR/HER2/PAM50); **cross-dataset drop 13–22%** (overall 0.82)
Selective routing (PubMed 41671586)	TCGA-BRCA, 924 PAM50 patients; CTransPath encoder + gated attention	**94.93%** accuracy; multimodal path invoked for only 38.2% (ambiguous) cases
HEALNet (NeurIPS 2024)	Heterogeneous biomedical data, 4 TCGA cohorts	Cross-attention fusion; explicitly handles **missing modalities**
DeepClinMed-PGM (2024)	TCGA + SYSMH cohorts	1-yr training AUC 0.979; external 1/2/3-yr AUC 0.851 / 0.878 / 0.938

The Open Problem. In-distribution performance is near-ceiling (0.96-0.99), but cross-cohort transfer loses **13–22% AUC**. The question is no longer *"can we fuse?"* but *"does the fusion survive contact with a different cohort?"*

Proposed Method & Expected Outcome



Experimental Plan.

1. Baseline: RNA-seq only
2. Baseline: WSI only (ResNet50 vs. foundation-model ablation)
3. Baseline: naive concatenation fusion
4. **Proposed**: cross-attention fusion, foundation-model encoder
5. TCGA → CPTAC generalization vs. reported 13–22% drop

Key Novelty. Swap M3FusionNet's generic ResNet50 for a pathology foundation model, targets the field's principal architectural weakness.

Expected Outcome. Per-biomarker AUC vs. published TCGA numbers; modality-contribution analysis; CPTAC generalization number; attention heatmaps linking receptor status to morphology.

Risks & Mitigations. Foundation-model weights gated (Hugging Face approval) → fallback to open-weight Virchow / CTransPath · PAM50 class imbalance (rare subtypes) → macro-averaged metrics, class weighting · Batch effects across TCGA sites → Macenko + ComBat · WSI storage/compute → precompute & cache frozen embeddings, fp16 on 24 GB RTX 6000